

LOCAL FOOD WASTAGE MANAGEMENT SYSTEM

End-to-End Data Analytics & Food Redistribution Platform

Project Type: Data Analytics Portfolio Project

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1. Executive Summary

Food wastage remains one of the most significant challenges affecting sustainability and food security. Restaurants, grocery stores, supermarkets, caterers, and households frequently dispose of edible surplus food while many individuals and organizations continue to face food shortages.

The Local Food Wastage Management System was developed as an end-to-end analytical platform that enables providers to list surplus food and allows receivers such as NGOs, shelters, and individuals to claim available food resources.

The solution combines database management, SQL analytics, and dashboard visualization to support data-driven decision-making and improve food redistribution efficiency.

Project Outcomes

KPI	Result
Total Food Listings	1,000
Total Claims	1,000
Total Food Units Available	25,794
Claim Completion Rate	33.9%
Database Tables	4
SQL Queries Implemented	15
Dashboard Tabs	6

The project successfully identifies supply-demand gaps, provider performance trends, and operational bottlenecks affecting successful food distribution.

2. Problem Statement

Large quantities of surplus food are wasted every day due to the absence of a centralized mechanism connecting food providers with potential receivers.

Key challenges include:

- Lack of visibility of available surplus food
- Inefficient communication between providers and receivers
- Food expiration before collection
- Geographic mismatch between supply and demand
- Limited monitoring of donation performance

The objective of this project is to create a centralized platform that enables efficient food redistribution while providing actionable insights through data analytics.

3. Business Objectives

The project was designed to achieve the following objectives:

Operational Objectives

- Create a centralized food listing system

- Enable food claiming and tracking
- Maintain provider and receiver information
- Support complete CRUD operations

Analytical Objectives

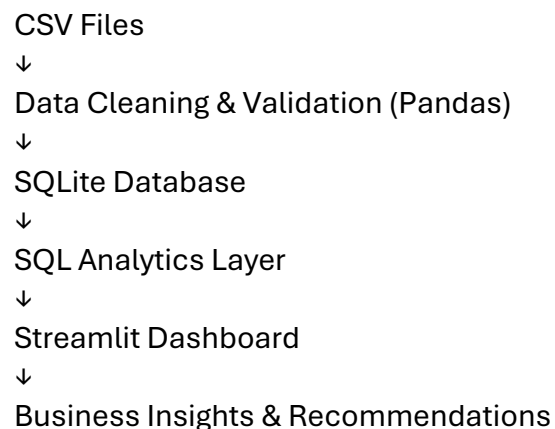
- Monitor food availability
- Measure claim completion rates
- Identify top-performing providers
- Detect supply-demand imbalances
- Generate business recommendations

4. Technology Stack

Layer	Technology
Programming Language	Python
Data Processing	Pandas
Database	SQLite
Query Language	SQL
Dashboard	Streamlit
Visualization	Streamlit Charts
Version Control	Git

5. System Architecture

The solution follows a layered architecture:



Purpose: Illustrates the complete flow from raw datasets to dashboard-driven insights.

6. Data Sources

The project utilizes four datasets.

Providers Dataset

Contains information about food providers including:

- Provider ID
- Provider Name
- Provider Type
- Address
- City
- Contact Information

Total Records: 1,000

Receivers Dataset

Contains information about food receivers including:

- Receiver ID
- Receiver Name
- Receiver Type
- City
- Contact Information

Total Records: 1,000

Food Listings Dataset

Contains information about available food including:

- Food ID
- Food Name
- Quantity
- Expiry Date
- Provider Information
- Food Type
- Meal Type
- Location

Total Records: 1,000

Claims Dataset

Contains claim transaction details including:

- Claim ID
- Food ID
- Receiver ID
- Claim Status
- Timestamp

Total Records: 1,000

7. Data Cleaning & Preparation

Data validation was performed before database creation.

Data Quality Checks

Validation Check	Result
Missing Values	0
Duplicate Records	0
Invalid Dates	0
Foreign Key Issues	0

Transformations Performed

- Converted Expiry_Date into DATE datatype
- Converted Timestamp into DATETIME datatype
- Verified referential integrity
- Standardized categorical values

8. Database Design

SQLite was used to create a normalized relational database.

Tables Created

1. Providers
2. Receivers
3. Food_Listings
4. Claims

Database Features

- Primary Keys
- Foreign Keys
- Indexed Columns
- Referential Integrity
- Optimized Query Performance

9. SQL Analysis

Fifteen SQL queries were developed to answer business-critical questions.

Query Categories

Provider Analysis

- Providers per city
- Provider type distribution
- Top providers

Receiver Analysis

- Receivers per city
- Receiver distribution

Food Analysis

- Food type trends
- Meal type trends
- Quantity availability

Claims Analysis

- Completion rate
- Pending claims
- Cancelled claims

Geographic Analysis

- Supply hotspots
- Demand hotspots
- City mismatch analysis

Figure 5: SQL Analytics Dashboard

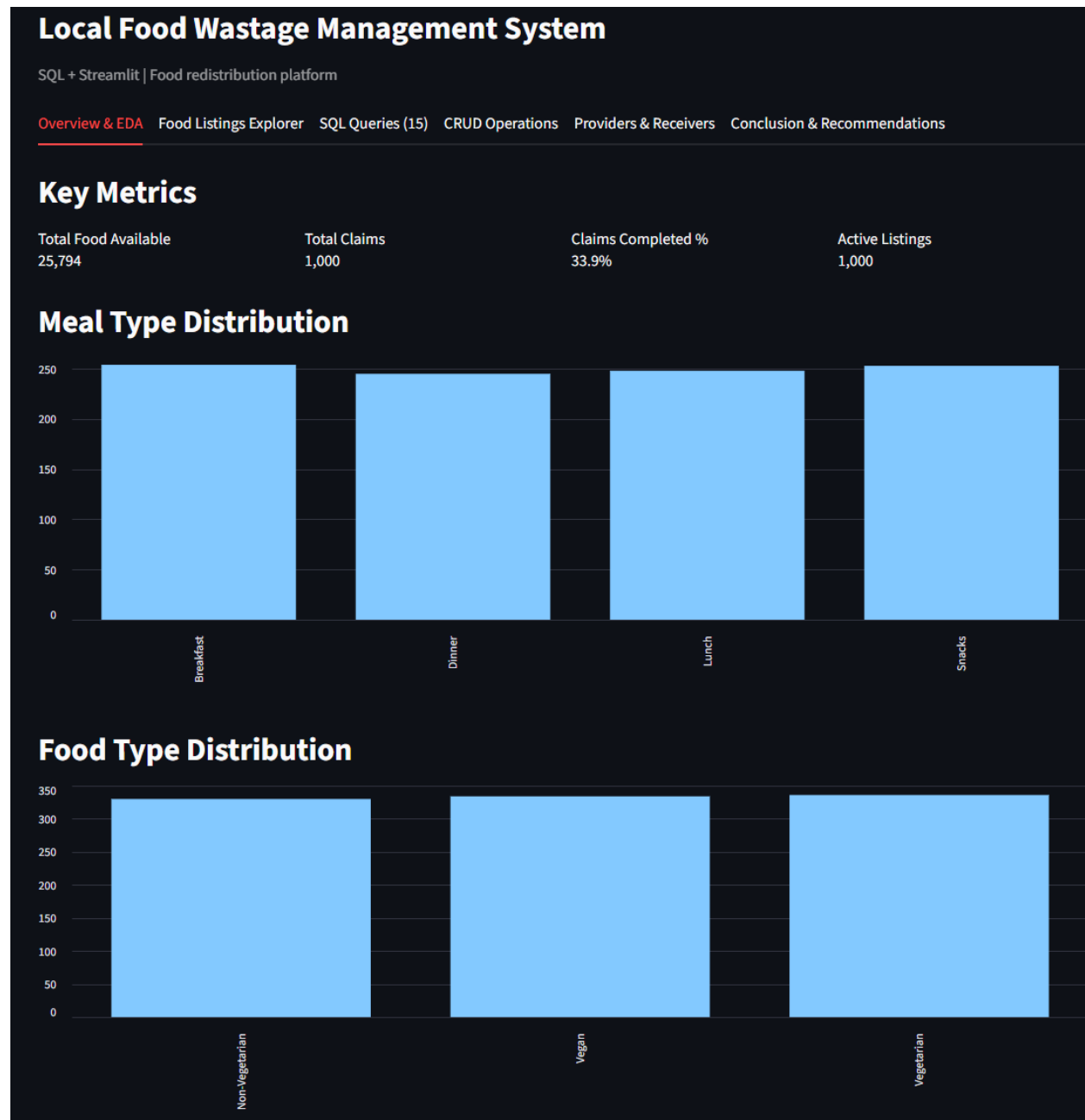
INSERT SCREENSHOT HERE

Caption: Figure 5. SQL query outputs displayed within the Streamlit dashboard.

10. Dashboard Development

A six-tab interactive dashboard was developed using Streamlit.

Tab 1: Overview & KPIs



Displays:

- Total food available
- Total claims

- Completion percentage
- Summary charts

Tab 2: Food Listings Explorer

Overview & EDAFood Listings ExplorerSQL Queries (15)CRUD OperationsProviders & ReceiversConclusion & Recommendations

Filter Food Listings

City / Location

Choose options

Food Type

Choose options

Meal Type

Choose options

Quantity Range

0

50

Food_ID	Food_Name	Quantity	Expiry_Date	Provider_ID	Provider_Type	Location	Food_Type	Meal_Type
1	Bread	43	2025-03-17 00:00:00.000000	110	Grocery Store	South Kellyville	Non-Vegetarian	Breakfast
2	Soup	22	2025-03-24 00:00:00.000000	791	Grocery Store	West James	Non-Vegetarian	Dinner
3	Fruits	46	2025-03-28 00:00:00.000000	478	Catering Service	Lake Regina	Vegan	Breakfast
4	Fruits	15	2025-03-16 00:00:00.000000	930	Restaurant	Kellytown	Vegan	Lunch
5	Soup	14	2025-03-19 00:00:00.000000	279	Restaurant	Garciaport	Vegan	Dinner
6	Vegetables	4	2025-03-24 00:00:00.000000	378	Supermarket	South Johnshire	Vegan	Snacks
7	Dairy	29	2025-03-25 00:00:00.000000	268	Grocery Store	Nolanmouth	Vegan	Dinner
8	Rice	49	2025-03-28 00:00:00.000000	343	Catering Service	South Christopherborough	Vegetarian	Lunch
9	Bread	46	2025-03-25 00:00:00.000000	47	Supermarket	East Andrewland	Non-Vegetarian	Snacks
10	Rice	41	2025-03-20 00:00:00.000000	528	Supermarket	Markport	Vegan	Dinner

Provides filtering by:

- City
- Food Type
- Meal Type
- Quantity

Tab 3: SQL Analytics

All 15 SQL Queries and Outputs (one after another)

1. Providers per city (top 10)

City	num_providers
New Carol	3
South Christopherborough	3
Bradleyport	2
Davidville	2
East Anthony	2
East Jennifer	2
East Lisa	2
East Melissa	2
East Stephanie	2
Jamesport	2

Returned 10 rows

2. Receivers per city (top 10)

City	num_receivers	
New Christopher	3	
Williamsfort	2	
West Richard	2	
West James	2	
Stevenmouth	2	
Spencermouth	2	
South Mark	2	
Smithshire	2	
Sarahaven	2	
Robertton	2	

Returned 10 rows

3. Provider type contributing most food (total quantity)

Provider_Type	total_quantity
Restaurant	6923
Supermarket	6696
Catering Service	6116
Grocery Store	6059

Returned 4 rows

4. Top 10 receivers by number of claims

Name	Type	City	claims_made
Scott Hunter	Individual	Greenton	5
William Frederick	NGO	Port Dean	5
Matthew Webb	Charity	West David	5
Anthony Garcia	Individual	Brownbury	5
Alvin West	Shelter	Kellybury	4
Jennifer Nelson	Individual	North Paul	4
Kristina Simpson	NGO	Melissaport	4
Kristine Martin	NGO	Thompsonhaven	4
Betty Reid	Charity	North Abigail	4
Chelsea Powell	Charity	Lake Rachelburgh	4

Returned 10 rows

5. Total food quantity available from all providers

total_available_quantity
25794

Returned 1 rows

6. Top 5 cities by number of food listings

City	num_listings
New Carol	6
South Kathryn	6
East Angela	5
Jimmyberg	5
Perezport	5

Returned 5 rows

7. Most commonly available food types

Food_Type	count
Vegetarian	336
Vegan	334
Non-Vegetarian	330

Returned 3 rows

8. Top 10 food items by number of claims

Food_Name	Food_Type	Meal_Type	times_claimed
Soup	Vegan	Lunch	5
Chicken	Vegan	Snacks	5
Fish	Vegan	Snacks	5
Rice	Vegan	Breakfast	4
Chicken	Vegetarian	Breakfast	4
Salad	Non-Vegetarian	Snacks	4
Soup	Vegetarian	Lunch	4
Dairy	Non-Vegetarian	Lunch	4
Fish	Non-Vegetarian	Breakfast	4
Vegetables	Vegetarian	Lunch	4

Returned 10 rows

9. Top 5 providers by successful (Completed) claims

Name	Type	City	successful_claims
Barry Group	Restaurant	South Kathryn	5
Miller Inc	Grocery Store	Coleburgh	4
Butler-Richardson	Grocery Store	East Heatherport	4
Barnes, Castro and Curtis	Restaurant	Zimmermanville	4
Harper, Blake and Alexander	Catering Service	Devinmouth	4

Returned 5 rows

10. Claim status percentages

Status	count	percentage
Completed	339	33.9
Cancelled	336	33.6
Pending	325	32.5

Returned 3 rows

11. Average claims per receiver (who made at least one claim)

avg_claims_per_active_receiver
1.6

Returned 1 rows

12. Most claimed meal type (by claim count)

Meal_Type	claims_count
Breakfast	278
Lunch	250
Snacks	240
Dinner	232

Returned 4 rows

13. Top 10 providers by total quantity donated/listed

Name	Type	City	total_quantity
Barry Group	Restaurant	South Kathryn	179
Evans, Wright and Mitchell	Catering Service	North Keith	158
Smith Group	Restaurant	Jimmyberg	150
Nelson LLC	Restaurant	Lake Andrewmouth	142
Ruiz-Oneal	Grocery Store	Lake Travis	140
Blankenship-Lewis	Catering Service	Jonathanstad	124
Campbell LLC	Supermarket	Perezport	123
Kelly-Ware	Catering Service	Bentleyburgh	123
Bradford-Martinez	Supermarket	New Carol	121
Shepherd and Sons	Catering Service	Pamelaberg	116

Returned 10 rows

14. Top 5 cities by total food quantity available

City	total_quantity
South Kathryn	179
Jonathanstad	169
New Carol	167
North Keith	158
Jimmyberg	150

Returned 5 rows

15. Top 5 cities by claim demand (number of claims on local food)

City	claims_from_city
East Heatherport	12
Jamesfurt	9
Lake Andrewmouth	9
East John	7
New Carol	7

Returned 5 rows

Tab 4: CRUD Operations

[Overview & EDA](#) [Food Listings Explorer](#) [SQL Queries \(15\)](#) [CRUD Operations](#) [Providers & Receivers](#) [Conclusion & Recommendations](#)

Add New Food Listing

Food Name	Provider Type
	Restaurant
Quantity	Location (City)
10 - +	South Kellyville
Expiry Date	Food Type
2025/04/01	Vegetarian ▾
Provider ID	Meal Type
1 - +	Breakfast ▾
Add Listing	

Update Claim Status

Claim ID
1 - +
New Status
Pending ▾
Update Status

Delete Food Listing (by Food_ID)

Food ID to delete
1 - +
Delete Listing

Supports:

- Create
- Read
- Update
- Delete

Tab 5: Provider & Receiver Directory

Overview & EDAFood Listings ExplorerSQL Queries (15)CRUD OperationsProviders & ReceiversConclusion & Recommendations

Providers (with contact for coordination)

Filter providers by city

All

Name	Type	City	Contact
Gonzales-Cochran	Supermarket	New Jessica	+1-600-220-0480
Nielsen, Johnson and Fuller	Grocery Store	East Sheena	+1-925-283-8901x6297
Miller-Black	Supermarket	Lake Jesusview	001-517-295-2206
Clark, Prince and Williams	Grocery Store	Mendezmouth	556.944.8935x401
Coleman-Farley	Grocery Store	Valentineside	193.714.6577
Lawson-Walters	Grocery Store	Shannonside	144-860-6074x60544
Ortiz-Lee	Restaurant	Lake Christopherburgh	(563)889-7190
Curtis-Lewis	Supermarket	Washingtonville	+1-521-027-8120x3301
Nguyen Inc	Supermarket	Tinamouth	898-609-2609x844
Hall, Nguyen and Martinez	Catering Service	West Cherylfort	(740)949-2480x40566

Receivers (with contact)

Filter receivers by city

All

Name	Type	City	Contact
Donald Gomez	Shelter	Port Carlburgh	(955)922-5295
Laurie Ramos	Individual	Lewisburgh	761.042.1570
Ashley Mckee	NGO	South Randalltown	691-023-0094x856
Erika Rose	NGO	South Shaneville	8296491111
John Romero	Individual	Bakerport	067.491.0154
Mandy Sutton PhD	Shelter	East Sharimouth	682-777-5357
Kenneth Baker	NGO	South Edwinborough	126-645-6386x4071
James Perez	Charity	Benjaminburgh	+1-986-748-6688
Emily Turner	Shelter	West Robert	001-631-022-6157x520
Mary Salazar	NGO	Wrightland	001-187-333-7518x11395

Displays contact information.

Tab 6: Recommendations

Overview & EDA	Food Listings Explorer	SQL Queries (15)	CRUD Operations	Providers & Receivers	Conclusion & Recommendations
Project Conclusion System built: Local Food Wastage Management System using Python, SQLite, and Streamlit. Data: 1,000 providers, 1,000 receivers, 1,000 food listings, 1,000 claims (0 nulls, clean). Total food available: 25,794 units across 1,000 active listings. Claim completion rate: 33.9% (majority pending/cancelled). Supply balance: Provider types ~23-26% each; Food types ~33% each; Meal types ~25% each. All 15 required SQL queries, filters, CRUD, and contact display are live and functional. Key Insights from the 15 Queries 1. Low conversion: Only 33.9% of claims reach Completed status despite 25,794 units available — coordination or timing gaps dominate. 2. Balanced but under-utilised supply: Even distribution across provider/food/meal types, yet claim demand concentrates in specific cities (queries 6, 14, 15). 3. Provider concentration: A small set of providers account for the highest successful claims and total quantity donated (queries 9 and 13). 4. Geographic mismatch: Top listing cities and top claim-demand cities are not perfectly aligned — surplus exists where demand is lower. 5. High pending/cancelled volume: ~66% of claims do not complete, directly reducing impact of the 1,000 listings. Recommendations 1. Targeted NGO partnerships in high-supply / low-claim cities Use query 14 + 15 outputs to prioritise outreach and collection routes in cities with high listed quantity but lower claim counts. Expected impact: lift completion rate 10-15 points within 60 days. 2. Recognition + priority for top successful providers Surface the top 5 providers from query 9 on the homepage and in the Listings Explorer. Expected impact: sustained high-quality supply and 20%+ more successful claims from the same cohort. 3. Automated expiry alerts + status nudges Add a nightly flag (Expiry_Date within 7 days) and in-app notifications before listings expire. Directly addresses the 33.9% completion bottleneck. 4. Increase receiver onboarding and claim support in top demand cities Focus new NGO/Individual registration drives and simplified claim flow in the 5 cities from query 15. Expected impact: higher claim volume and reduced pending queue. 5. Weekly dashboard review ritual Run the full 15-query set every Monday; track completion % and top-city mismatch as primary KPIs. Use the CRUD forms to correct stale listings in real time.					

Displays insights and actions.

11. Exploratory Data Analysis

Several visualizations were developed to understand data distributions.

Food Type Distribution

Analysis showed balanced distribution across food categories.

12. Key Findings

Finding 1

A total of 25,794 food units remain available across active listings.

Finding 2

Only 33.9% of claims reach successful completion.

Finding 3

Approximately 66% of claims remain pending or cancelled.

Finding 4

Supply remains balanced across provider and food categories.

Finding 5

Demand is concentrated in selected cities.

Finding 6

A small group of providers contributes a majority of successful donations.

13. Business Recommendations

Recommendation 1

Target NGO and shelter onboarding initiatives in high-supply and low-demand cities.

Expected Impact: Increase claim completion rate by 10–15%.

Recommendation 2

Implement automated expiry notifications 48–72 hours before expiration.

Expected Impact: Reduce food wastage due to expiration.

Recommendation 3

Introduce provider recognition and incentive programs.

Expected Impact: Increase participation and successful donations.

Recommendation 4

Improve receiver onboarding in high-demand cities.

Recommendation 5

Conduct weekly KPI reviews using dashboard analytics.

14. Challenges Encountered

The following challenges were addressed during development:

- Managing relational database constraints
- Building efficient SQL joins
- Handling date-based calculations
- Designing responsive dashboard layouts
- Ensuring data consistency across modules

15. Future Enhancements

Potential future improvements include:

- Mobile application
- Email and SMS notifications
- Real-time food availability tracking
- Google Maps integration
- Machine Learning demand forecasting
- Route optimization for food pickup
- Provider performance scoring

16. Skills Demonstrated

SQL

- Joins
- Aggregations
- Group By
- Indexing

- CRUD Operations

Python

- Pandas
- Data Cleaning
- ETL Development

Data Analytics

- KPI Design
- Business Analysis
- Data Interpretation

Dashboarding

- Streamlit
- Interactive Filters
- Visualization

Database Management

- SQLite
- Schema Design
- Referential Integrity

17. Conclusion

The Local Food Wastage Management System successfully demonstrates the complete lifecycle of a modern data analytics solution.

The project integrates data preparation, relational database design, SQL analytics, dashboard development, and business intelligence into a single platform.

By identifying a claim completion rate of only 33.9% and uncovering significant geographic mismatches between supply and demand, the project provides actionable insights that can help organizations reduce food wastage and improve food distribution efficiency.

The solution showcases practical skills in SQL, Python, data analytics, database management, and dashboard development while addressing a real-world social impact problem.